2 Digital literacy: Searching the world wide web

You will learn

- how to find information on the web and describe the sources you use
- how to choose information you find on the web, and give reasons for your choices
- how web search engines select and show useful information.

Last year you learned how to use a search engine to find information on the web. Some of the information you find on the web will be correct and reliable. Some of it will be incorrect and misleading. In this unit you will learn how to choose the best information for you.



Talk about...

How do you learn about what is happening in the world? Do you use the web to find news? Can you trust what you read and watch on the web?



Learning outcomes: Obtain information from online sources and describe the sources used; Choose information from online sources, and give reasons for choices; Explain how online searches select and show useful information



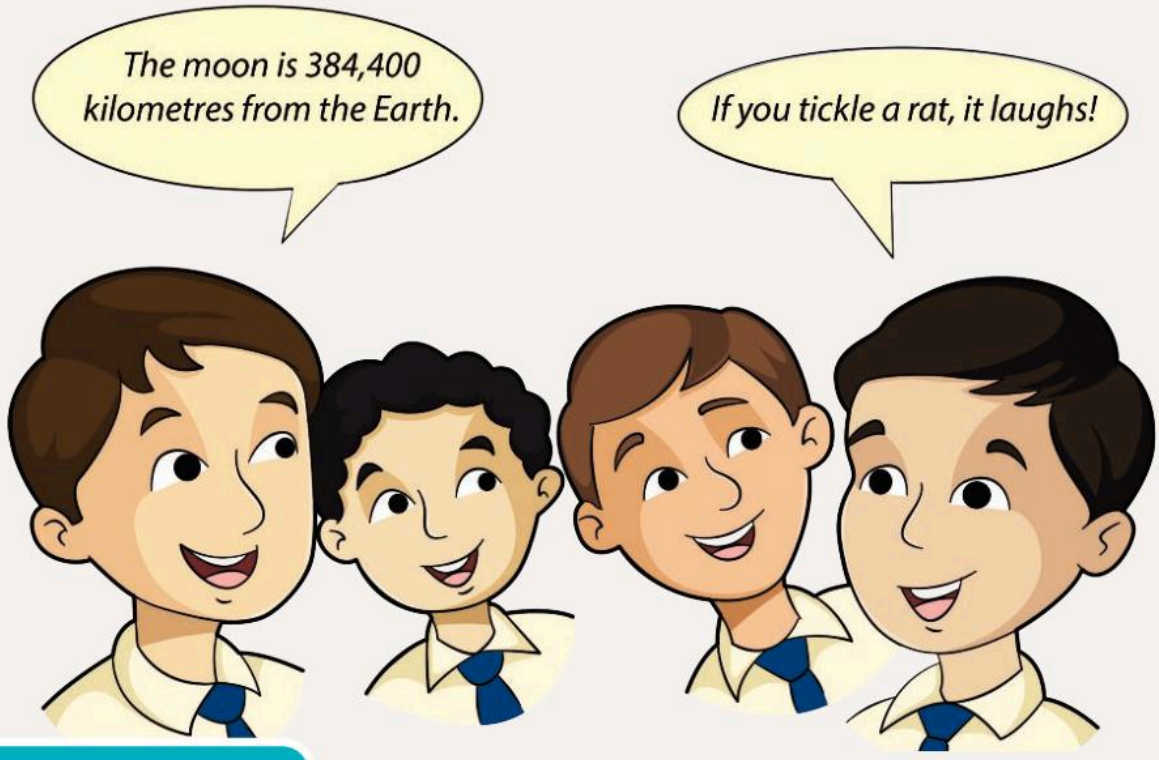
Class activity

key word
search engine bookmark
spider web crawler meta tag
the web web page website
copyright

What is the most interesting fact that you have learned this week? Your fact can be funny. It can be sad. It can be about a big world event or about something that happened in your town. How did you find out this fact?

Discuss your facts in a small group. Make a list of the facts that your group thinks are the most interesting. Make sure you include one fact from each person in the group.

Do you believe all of the facts are true? Share your findings with the rest of the class.



Did you know?

The world's first **search engine** was invented in 1990 by Alan Emtage. Alan Emtage was born in Barbados but moved to Montreal, Canada, to study at McGill University. His job at the university was to help other students find articles on the internet. He wrote a search engine to save himself time. He called his search engine Archie.

2.1 Searching the web

In this lesson

You will learn:

- four golden rules to help you to search the web.

Spiral back

Last year you learned how to carry out basic world wide web searches. You learned how to use a search engine to find information quickly on the web. In this lesson you will review the search techniques that you learned last year.

Four golden rules

1 Choose the right key words

Make sure you understand the question you need to answer. Before you enter anything into your search engine:

- Make a list of **key words**.
- Underline the most important key words.

Make sure that the question you type in the search engine contains the important key words. Your question does not need to include punctuation or short joining words like 'is' and 'the'.

2 Choose the right search engine

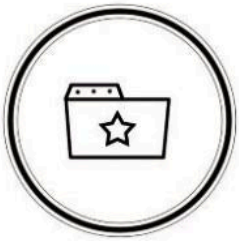
There are many search engines that you can use. Google, Bing and Yahoo are examples. Try using different search engines. Find out which one works best for you.

Some search engines are designed for young people. The advantages of using a child-friendly search engine are:

- The links are easier for you to read and understand.
- You are less likely to see unsuitable web pages.
- There are fewer adverts.



3 Bookmark your favourite sites



Adding a **web page** to your **bookmarks** makes it easy to find next time you want to use it. You don't have to search for the page again. You click on the link in your bookmarks list and go straight to the page.

Your bookmarks can be a good place to start a search. **Websites** you have found useful in the past might provide the information you need.

4 Try again

Sometimes a search does not give you the information you need. If that happens, try again.

- Try to think of different key words.
- Will a different search engine give better results?
- Will any of your bookmarked pages lead you to helpful information?



Activity

Use a search engine to answer these questions.

1. What is Tungurahua, and where is it?
2. What happened at Tungurahua in March 2016?
3. Find a picture of Tungurahua.
4. Find one interesting fact about Tungurahua.
5. How do you pronounce Tungurahua?

Think again

Work in a small group. Discuss websites that you have found useful in your studies. Make a list of the best websites you have found. Why do you like them? What subjects do these websites help with? Present your list to the class.



Extra challenge

Search the web to find the name of this bird.

Hint: To get a list of key words, type the things you see in the picture, such as 'black head'. What else do you see? Do an image search and find a picture that matches this one.

Which combination of key words worked best?



2.2 How search engines work

In this lesson

You will learn:

- how a search engine finds web pages to answer your search questions.



How large is the web?

At the start of 2019 there were 1.9 billion (1,900,000,000) websites on the web. It is easy to imagine what a small number like 10 or 100 looks like. It is more difficult to imagine what a large number like 1 billion looks like.

Think about this book:

- This lesson contains 500 words.
- The complete book contains 25,000 words.
- A pile of 76,000 books would contain 1.9 billion words. That is the same as the number of websites on the web.
- The pile of books would be 400 metres tall.
- That is half the height of the world's tallest building, Burj Khalifa.

How does a search engine work?

When you enter a web search, your question is sent over the internet to a search engine. The search engine uses a powerful computer to look for matches to your key words. A list of links is displayed on your screen.

A search engine would take a very long time to search all 1.9 billion websites on the web. To be able to send you a list of links quickly, the search engine must work in a different way.

Spiders and web crawlers

A search engine uses **web crawling** to search the web. The search engine sends a piece of software called a **spider** to crawl around the web and collect information.

When a spider reaches a web page, it records every word on the page and counts how many times each word is used. Then the spider follows links on the web page and carries out a word count on each new page it finds. Eventually, the spider will visit and record every web page on the web.



Search engine index

The spider software sends the information it collects back to the search engine. The information is stored in a special list called an **index**. When you enter a search question, the search engine searches its index. The index is very large, but searching it is much quicker than searching the whole web.

A search engine compares each key word in your search question to the words stored in its index. It ignores short joining words like 'the', 'what' and 'why'. These words appear on every web page so won't help in a search.

The search engine uses an **algorithm** to compare websites in its index with the key words in your web search. The algorithm chooses the web pages to answer your question.



Activity

Copy this search question: 'What is the longest river in Africa?'

Draw circles around all the words that the search engine will ignore. What is the answer to the question?



Extra challenge

A search engine uses an algorithm to choose websites to answer a web search question. What is an algorithm? Use the knowledge you have already gained in this course. Search the web for more information if you need to.

Think again

Carry out a web search to find a video or animation that explains how a search engine works. Watch the video and make notes. Bookmark the video that you find most useful.

2.3 Search results

In this lesson

You will learn:

- how a search engine shows the links it finds.

How many pages?

The web contains 1.9 billion websites. Whatever question you ask, a search engine will find many matching pages.

When a search engine shows a list of links it includes the total number of pages it found. The search shown in this image has found 722,000,000 results.



You can make the list shorter by adding more key words to a search. However, you do not usually need to do this.

How a search engine sorts results

A search engine tries to put the most useful links at the top of the list of results. It uses several methods.

- **How popular is a page?** The most popular web pages are shown on the first page of search results. A page is popular if many other sites link to it.
- **How up to date is a page?** Web pages that change often are shown near the top of the search results. Adding new information to a web page is called updating.
- **Can the website be trusted?** Websites that a search engine trusts are shown near the top of the results. This does not mean that you can believe everything you see on websites near the top of the list. You still need to check the facts.

- **Which pages have you looked at before?** A web browser keeps a list of websites you have looked at. The search engine uses this information to predict which websites you will want to see in the future.

Adverts

Most search engines show adverts at the top of the list of results. The owners of these websites pay for the adverts. Adverts are not always relevant to your search.



Meta tags

The owner of a website wants the world to see their pages. One way they can do this is to use **meta tags**. Meta tags are like key words. They are single words that describe a web page.

Web developers add meta tags to the pages they create. People who read a web page cannot see the meta tags, but search engines can see them. If any key words in your search question match the meta tags on a web page, the search engine shows that page near the top of the results list.



Activity

Carry out another web search about Tungurahua. How many results does the search return? Add some more key words to your search. What happens to the number of results?



Extra challenge

How many adverts are shown when you search for Tungurahua? What are the adverts for? Do the adverts give you any useful information?

Think again

Carry out a web search for more information about how search engines work. How many adverts are at the top of the list of results? How many of the adverts give you useful and relevant information?

2.4 Choosing web content

In this lesson

You will learn:

- how to choose information carefully from the world wide web.

Can you trust web content?

A web search will provide you with links to many web pages. The links lead to information you can use to complete a project. How will you decide which web pages to use and which to reject?

Sometimes content on web pages is wrong. The person who wrote the content may have made a mistake or the information may be out of date. Sometimes the writer of a web page may deliberately try to mislead you.

Checklist for choosing web content

Ask the questions in this checklist to help you to choose content that is correct, reliable and relevant.

1 Who wrote the content?

Who owns the website? Websites that belong to governments, charities, universities or other large organisations are usually reliable sources of information.

Who wrote the web page? It is a good sign if the writer has included their name on the web page. Sometimes web pages also include the writer's job title and contact details.



2 Is the content up to date?

Can you find the date when the web page was last updated? Having up-to-date information is important. It is very important in computer studies because technology changes quickly. It can be less important for subjects like history and mathematics.

3 Is the page appropriate for your age?

Content: Can you understand the content or is it too complicated for you? Is there more detail than you need? Is the content too simple for you?

Language: Are there too many words that you don't understand? Are the sentences too long? This is especially important if you are reading content in a second language.

4 Is the content relevant?

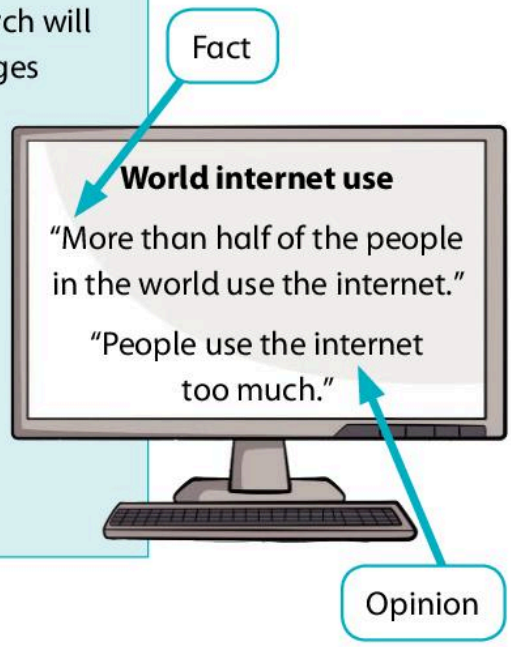
Does the page answer your search question? A web search will find many interesting and well-written pages. Which pages provide the information you need?

5 Does the page give you facts or opinions?

Facts: A fact is something that you can check to find out if it is true.

Opinions: An opinion is a what someone thinks or feels.

Does the page give you facts to help you to make up your own mind? Or is the writer giving their opinion to try to persuade you to agree with them?





Activity

Choose a web page you have visited during this unit. Use the checklist in this lesson to write a report about the page.



Explore more

Ask a member of your family to suggest a web page you can both look at. Working together, complete the checklist shown on this page. What does your family member think of the website? Good or bad?



Extra challenge

Search the web for information about the impacts of climate change. Find a page that you like. Is the page mainly facts or opinions? Give one example of each.

2.5 Giving credit

In this lesson

You will learn:

- what copyright is
- how to give credit to another person's work.

What is copyright?

If you create something, you have **copyright**. Copyright means that you own the work you have created.

Copyright protects the person who created the work. Copyright means that other people:

- cannot copy that work
- must ask for the copyright owner's permission to use or share the work
- must say who created the work.

Using other people's work in your school work

- Make sure you have permission. You can usually use small amounts of other people's work in projects and assignments without asking. If you are not sure, check with your teacher.
- Give credit to the person who created the work.

Giving credit for work you use

If you use part of someone else's work, you must say who created the work. Your credit should include these four pieces of information:

Zheng Ying, Saving the Whales, thedailynews.com, October 2019

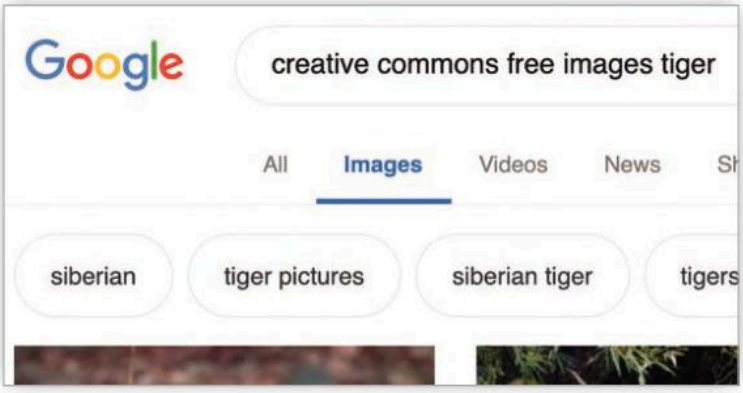
- Author
- Title of the article or web page
- Website
- Date the work was created or the date you accessed the website

You may not be able to find all four pieces of information. Use all the information you can find.

Using photos in your work

Photos can make your projects look special. A photo can explain an idea more clearly than words alone.

Some photographers and artists allow people to use their work free of charge. They use a creative commons licence. To find creative commons images, use a web search such as 'creative commons free images tiger'.



Click on the 'Images' tab in your search engine to see photos that you can use. Remember to include a credit.



Plagiarism

It is good to use information you have found on the web in your school work. Quotes and facts from experts will improve your work. Pictures will add impact to your projects and assignments.

However, never pretend that another person's work is your own. That is called **plagiarism**. Always credit another person's work. Be confident! You can create great work yourself.



Activity

In Lesson 2.4 you found a web page about the impacts of climate change. Go back to the page and find a piece of text you would like to quote. Copy and paste the quote into a document. Add a credit at the end of the quote.



Extra challenge

Find a creative commons photo that illustrates the quote you used in the activity. Add the photo to your document. Add a credit for the photo.

Think again

Why does crediting another person's work matter? If it is on the internet, why not just use it? How would you feel if someone copied your work and claimed it was their own?

2.6 Web challenge

In this lesson

You will:

- use the skills and knowledge you have learned in this unit to complete a team web challenge.

The task

Work in a team to create a set of fact sheets. Choose the topic for your fact sheets. Here is an example.

Big cat fact sheets: Each fact sheet will be about one big cat (for example, lion, puma, tiger, cheetah, leopard, jaguar). The fact sheets will all have the same design and the same headings.

Here are some more ideas: famous scientists, football teams, popular singers, capital cities, endangered species.

BIG CATS FACT SHEET: AFRICAN LION



Fun facts

- Lions sleep for 20 hours a day.
- You can hear a lion's roar from 8 km away.
- A lion cannot purr.

Facts	
Home	Southern and eastern Africa
Habitat	Grasslands
Food	A lion eats large animals such as zebra, antelope and wildebeest. Lionesses do most of the hunting.
Families	Lions live in groups called prides. Young lions are called cubs.
Physical features	Male lions have a mane. A male lion weighs 180 kg. A female weighs 150 kg. The lion is the second largest big cat.
Speed	A lion can run at 80 kph over short distances.
Endangered?	Lions are a vulnerable species. Their numbers are declining.

1 Plan your work

Before you start the challenge, plan how your team will work.

- What topic will you choose?
- How will you divide the work between the team members?
- How will you design your fact sheets? What headings will you use? Where will you put photos?

2 Choose websites

Before you start to collect information, each team member should carry out a web search and choose a relevant website. Then decide together which websites you will use to collect information.



3 Collect your facts

Research the information you need. Copy and paste the information into a blank word processor document. You will put it into your fact sheet design later.

4 Create your fact sheets

When you have collected enough information, transfer the information into a fact sheet. Remember to work as a team to make sure your fact sheets all look the same.



Activity

Working in a team, divide up the work you have to do. Create several fact sheets about your chosen topic.



Extra challenge

Discuss the web challenge with your team. Was it a success? What websites have been particularly helpful? Were any of the topics too hard or too easy?



Be creative

Make your fact sheets look interesting and professional. Which fonts and heading sizes will you use. Which colour combinations look best? Try looking at some fact sheets online for design ideas.

Think again

Make a list of the websites you used for this task. Rate how useful each one is.

Check what you know

You have learned

- how to find information on the web and describe the sources you use
- how to choose information you find on the web, and give reasons for your choices
- how web search engines select and show useful information.

Test

Look at the picture and answer the questions.



1. Which words are the important key words in this search question?

   'What is the fastest car in the world?'

2. The picture shows a search engine in a web browser. What is a search engine?
3. Why is it important to check the date of information you find on the web?
4. What does a web crawler do?
5. A search engine uses different methods to make sure relevant websites are near the top of the list of search results. State three of these methods.



Activities

Carry out a web search to find information about the wildlife in China. Look at some of the websites in the search engine's list of results. Choose the website you think is best for you to find out information. Then answer the questions below.

1 List the key words that you used in your web search.

2 Write down two pieces of information about wildlife in China that you found in your search.

3 List three websites you thought were useful.

4 Which website did you choose as the best?

5 Give reasons why you chose the website. Give as much information as you can about the website you chose.

Self-evaluation

- I answered test questions 1 and 2.
- I completed activities 1 and 2. I carried out a web search using key words. I found out some information about wildlife in China.
- I answered test questions 1–4.
- I completed activities 1–4. I carried out a web search and found three useful websites. I chose the best website to find out about wildlife in China.
- I answered all the test questions.
- I completed all the activities.

Re-read any parts of the unit you feel unsure about. Try the test and activity again – can you do more this time?